

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409454
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ung; Thang et al.

Evaporative control lid for multi-well sample trays

Abstract

A cover configured to reduce evaporation from a multi-well sample tray while permitting sampling of material housed in separate wells of the tray. The cover may directly engage the sample tray and move with the sample tray or the cover may be stationary while permitting movement of the sample tray relative to the cover.

Inventors:	Ung; Thang (San Ramon, CA), Crawford; Colin (Oakland, CA)
Applicant:	SARTORIUS BIOANALYTICAL INSTRUMENTS, INC. (Bohemia, NY)
Family ID:	86272329
Assignee:	Sartorius BioAnalytical Instruments, Inc. (Bohemia, NY)
Appl. No.:	17/721877
Filed:	April 15, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230330678 A1	Oct. 19, 2023

Publication Classification

Int. Cl.: B01L3/00 (20060101)

U.S. Cl.:

CPC B01L3/50853 (20130101); B01L2300/046 (20130101); B01L2300/0829 (20130101)

Field of Classification Search

CPC: B01L (3/50853); B01L (2300/046); B01L (2300/0829); B01L (2200/025); B01L (2200/026); B01L (2200/142); B01L (2300/04); B01L (2300/041)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5604130	12/1996	Warner et al.	N/A	N/A
5972694	12/1998	Mathus	N/A	N/A
7152492	12/2005	Zwick et al.	N/A	N/A
7309603	12/2006	Ma et al.	N/A	N/A
7854343	12/2009	Ellson et al.	N/A	N/A
10625264	12/2019	Hong et al.	N/A	N/A
2004/0089615	12/2003	Weiss	210/739	B01L 3/50853
2004/0115798	12/2003	Ma et al.	N/A	N/A
2006/0188404	12/2005	Gjerde	N/A	N/A
2010/0197525	12/2009	Davies et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 02/072423	12/2001	WO	N/A
2016138338	12/2015	WO	N/A

OTHER PUBLICATIONS

Wagener et al., Eppendorf 96-Well Cell Culture Plate—A Simple method of minimizing the edge effect in cell-based assays. Eppendorf. Apr. 2014;326: 6 pages. cited by applicant

Primary Examiner: Wecker; Jennifer

Assistant Examiner: Castaneda; Steven Ray

Attorney, Agent or Firm: Wolf, Greenfield & Sacks, P.C.

Background/Summary

BACKGROUND

(1) Bio-layer interferometry (BLI) is an analytical technique commonly used to measure biomolecular interactions. BLI analysis commonly uses a multi-well sample tray with each well containing a biomolecule in a suitable liquid. A typical multi-well sample tray 2 has a plurality of regularly spaced sample wells 4 arranged in a rectangular configuration. In most configurations, sample tray 2 rests on a shaker or other device which provides movement sufficient to maintain the material 6 within sample wells 4 in the form of a suspension. Due to the continuous movement of sample tray 2 and the small sample size, evaporative loss of liquid material 6 from sample wells 4 sometimes occurs leading to a decrease in accuracy of the BLI analysis. See for example FIG. 10. Therefore, a multi-well sample tray evaporation control cover which precludes or limits evaporative loss from the sample trays 2 will enhance the accuracy of the BLI analysis. Preferably, the evaporation control cover will achieve this goal while permitting BLI analysis without removal of the evaporation control cover and while permitting continuous movement of the multi-well

sample tray.

SUMMARY

(2) In one embodiment the present invention provides an evaporation control cover for use with a multi-well sample tray. The multi-well sample tray has a plurality of regularly spaced sample wells as defined by an outer perimeter of sample wells, with additional sample wells located within the outer perimeter of sample wells. The evaporation control cover comprises a plurality of sample holes arranged to correspond to the plurality of sample wells. The sample holes are defined by an outer perimeter of holes with additional sample holes located within the outer perimeter of holes. Optionally, a fluid port located in the cover provides fluid communication through the cover. The sample holes located within the outer perimeter of holes have a first diameter while the sample holes forming the perimeter holes have a second diameter which is less than or equal to the first diameter. The evaporation control cover carries a downwardly projecting flange configured to fit over the multi-well tray.

(3) In an alternative embodiment, the present invention provides an evaporation control cover for use with a multi-well sample tray. The multi-well sample tray has a plurality of regularly spaced sample wells defined by an outer perimeter of sample wells with additional sample wells located within said outer perimeter of sample wells. The evaporation control cover comprises a top. The top includes a plurality of sample holes corresponding to the plurality of sample wells.

Additionally, the top carries a downwardly projecting flange configured to fit over the multi-well tray. The cover also includes a bottom. The bottom has a plurality of upwardly projecting ports providing fluid communication through the bottom. The upwardly projecting ports configured to correspond to said plurality of sample wells. The bottom carries an upwardly projecting flange configured to fit within the downwardly projecting flange of the top. The upwardly projecting flange and the upwardly projecting ports define a reservoir suitable for retaining a liquid. Further, the upwardly projecting ports provide fluid communication between the sample holes and the sample wells. This embodiment may optionally include a wettable insert capable of absorbing and releasing a liquid. The wettable insert has a plurality of insert holes corresponding to said plurality of sample wells.

(4) In another alternative embodiment, the present invention provides an evaporation control cover for use with a multi-well sample tray. The multi-well sample tray has a plurality of regularly spaced sample wells arranged in a rectangular configuration as defined by a first outer perimeter row, a second outer perimeter row, a third outer perimeter row and a fourth outer perimeter row. The outer perimeter rows defining the rectangular configuration have a first corner well, a second corner well, a third corner well and a fourth corner well with additional rows of sample wells located within the rectangular configuration. The evaporation control cover comprises a top. The top includes a plurality of sample holes arranged in a rectangular configuration corresponding to the plurality of sample wells and defined by a first top outer perimeter row, a second top outer perimeter row, a third top outer perimeter row and a fourth top outer perimeter row. The top outer perimeter rows defining the rectangular configuration further include a first top corner hole, a second top corner hole, a third top corner hole and a fourth top corner hole with additional rows of sample holes located within the rectangular configuration. The sample holes located to the interior of the first top outer perimeter row, the second top outer perimeter row, the third top outer perimeter row and the fourth top outer perimeter row have a first diameter. The sample holes within the first top outer perimeter row, the second top outer perimeter row, the third top outer perimeter row and the fourth top outer perimeter row have a second diameter which is less than or equal to the first diameter. Additionally, the top carries a downwardly projecting flange configured to fit over the multi-well tray. The cover also includes a bottom. The bottom has a plurality of upwardly projecting ports providing fluid communication through the bottom. The upwardly projecting ports have a rectangular configuration corresponding to the plurality of sample wells. The rectangular configuration is defined by a first bottom outer perimeter row, a second bottom outer perimeter

row, a third bottom outer perimeter row and a fourth bottom outer perimeter row. The bottom outer perimeter rows defining the rectangular configuration have a first corner upwardly projecting port, a second corner upwardly projecting port, a third corner upwardly projecting port and a fourth corner upwardly projecting port. Additional rows of upwardly projecting ports are located within the rectangular configuration. The bottom carries an upwardly projecting flange configured to fit within the downwardly projecting flange of the top. The upwardly projecting flange and the upwardly projecting ports define a reservoir suitable for retaining a liquid. Further, the upwardly projecting ports provide fluid communication between the sample holes and the sample wells. This embodiment may optionally include a wettable insert capable of absorbing and releasing a liquid. The wettable insert has a plurality of insert holes in a rectangular configuration corresponding to said plurality of sample wells.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 provides a perspective view of one embodiment of the multi-well sample tray evaporation control cover.
- (2) FIG. 2 provides an exploded view of the evaporation control cover of FIG. 1.
- (3) FIG. 3 provides a cut-away view along line 3-3 of FIG. 1 showing gap A and the multiple layers making up the evaporation control cover of FIGS. 1 and 2.
- (4) FIG. 4 provides a partial cut-away perspective view depicting an embodiment of the evaporation control cover as installed on a multi-well tray where the evaporation control cover moves with the multi-well tray.
- (5) FIG. 5 provides a partial cut-away perspective view depicting another embodiment of the evaporation control cover where the evaporation control cover remains stationary while the multi-well tray moves beneath the evaporation control cover.
- (6) FIG. 6 depicts an analytical probe passing through the embodiment of FIG. 4 into a sample well.
- (7) FIG. 7 depicts an analytical probe passing through another embodiment of the evaporation control cover where the evaporation control cover lacks a bottom and a wettable insert, and the evaporation control cover remains stationary.
- (8) FIG. 8 depicts a partial cut-away perspective view depicting the embodiment of FIG. 7.
- (9) FIG. 9 depicts a partial cut-away perspective view depicting another embodiment of the evaporation control cover as installed on a multi-well tray where the evaporation control cover moves with the multi-well tray.
- (10) FIG. 10 depicts a prior art view of an analytical probe positioned within a well of a multi-well tray lacking an evaporation control cover.

DETAILED DESCRIPTION

(11) The drawings included with this application illustrate certain aspects of the embodiments described herein. However, the drawings should not be viewed as exclusive embodiments. For simplicity and clarity of illustration, where appropriate, reference numerals may be repeated among the different figures to indicate corresponding or analogous elements and the drawings are not necessarily to scale. Throughout this disclosure, the terms “about”, “approximate”, and variations thereof, are used to indicate that a value includes the inherent variation or error for the device, system, the method being employed to determine the value, or the variation that exists among the study subjects. Finally, the description is not to be considered as limiting the scope of the embodiments described herein.

(12) FIGS. 1-9 depict embodiments of the evaporation control cover 10. As depicted, evaporation control cover 10 is particularly adapted for use with a multi-well sample tray 2 having sample wells

4. A typical multi-well sample tray 2 has 96 sample wells 4. Of course, the configuration of evaporation control cover 10 may be modified to accommodate multi-well sample trays 2 of differing configurations.

(13) As depicted in FIG. 4, evaporation control cover 10 may be configured to engage sample tray 2 and thereby move with sample tray 2. In one embodiment, the design of evaporation control cover 10 may provide a close engagement with sample tray 2. For example, evaporative control cover 10 may be configured to provide a frictional or snap fit securement to sample tray 2. FIGS. 6 and 9 depict examples where cover 10 nests over and engages sample tray 2. Typically, when evaporation control cover 10 engages sample tray 2, holes 22 remain in a consistent aligned position over sample wells 4.

(14) Alternatively, evaporation control cover 10 may be configured to permit movement of sample tray 2 while evaporation control cover 10 remains stationary. As depicted in FIGS. 5, 7 and 8 evaporation control cover 10 may be configured to permit securement of evaporation control cover 10 to a surface outside of the region supporting multi-well sample tray 2. In this configuration, evaporation control cover 10 may touch the upper surface of multi-well sample tray 2 so long as the contact does not inhibit the movement of multi-well sample tray 2. More typically, a slight gap 9 sufficient to permit movement of the multi-well sample tray 2 relative to control cover 10 will be provided between the upper surface 8 of multi-well sample tray 2 and the lower surface 46 of evaporation control cover 10. Typically, the gap will be between about 0.1 mm and about 1.0 mm. Thus, gaps of 0.2 mm, 0.3 mm, 0.4 mm, 0.5 mm, 0.6 mm, 0.7 mm, 0.8 mm, and 0.9 mm will also be appropriate.

(15) One embodiment of evaporation control cover 10 will be described with reference to FIGS. 1-6. As depicted therein, evaporation control cover 10 includes a top 20, a bottom 40 with a wettable insert 30 retained between top 20 and bottom 40. The combination of top 20, wettable insert 30, and bottom 40 cooperate to reduce or preclude the loss of liquid from sample wells 4.

(16) Without wishing to be bound by theory, the addition of liquids to wettable insert 30 is believed to enhance the functionality of evaporative control cover 10. Wettable insert 30 may have a thickness greater than Distance A depicted in FIG. 3. Distance A corresponds to the gap between the upper surface of bottom 40 and the lower surface of top 20. Thus, in some embodiments wettable insert 30 may be compressed between top 20 and bottom 40, i.e., insert 30 has a thickness greater than Distance A. Typically, wettable insert 30 will have a thickness which is 1.59 mm. However, wettable insert 30 may have a thickness which is less than, equal to or greater than distance A as depicted in FIG. 3. Depending on the application of evaporative control cover 10, distance A may range from about 0.1 mm to about 2 mm, from about 0.25 mm to about 1.5 mm or from about 0.5 mm to about 1 mm. Wettable insert 30 may be prepared as a felt, a non-woven or a woven material from a wide variety of materials capable of holding a liquid. For example, felts may be prepared from polypropylenes and polyesters. Sponges prepared from silicone, polyester, polypropylene, and polyethylene are also suitable for use as wettable insert 30. Likewise, open-cell foams prepared from silicone, polyurethane or polyethylene are suitable. Alternatively, a blanket like material prepared from polyimides will suffice.

(17) As depicted in FIGS. 1-2, 4-5, 8, and 9 cover 10 includes an optional fluid port 24. In the depicted embodiment of FIGS. 1-2 4-5, 8, and 9 fluid port 24 provides fluid communication through top 20 to the interior of evaporation control cover 10 including wettable insert 30 and bottom 40. However, port 24 may take other forms. For example, as depicted in FIGS. 1, 4, 6, and 9 an alternative opening 26 through top 20 allows access to either wettable insert 30 or reservoir 48 through which fluid may be added. Alternative opening 26 may be located at any convenient location on a side of top 20. While FIG. 1 depicts both opening 26 and port 24, typically only one of these two elements will be present. In most instances the liquid used to wet wettable insert 30 will be the same as the liquid used within sample wells 4 less the biological material being analyzed. However, any liquid which will not interfere with the analytical process, and which will

produce a sufficient partial pressure above sample wells **4** may be used. Typically, the fluid used to wet wettable insert **30** will be added to evaporation control cover **10** through fluid port **24**. Of course, prior to initial assembly of evaporation control cover **10**, wettable insert **30** may be pre-wetted with the desired fluid.

(18) The wettable insert **30** may be pre-wetted through a variety of techniques, such as by applying a seal (e.g., film) over the top **20** and/or bottom **40** of the cover **10** to contain the wettable insert **30**; sealing the cover **10** with the wettable insert **30** in a bag; and/or positioning one or more plugs around the holes **32** of the wettable insert **30**. The plugs may be made from a variety of materials, such as elastomer or plastic. Other techniques may also be used in other embodiments for pre-wetting the wettable insert **30** with the desired fluid.

(19) In an alternative embodiment, wettable insert **30** may be replaced with any suitable solution used to wet wettable insert **30**. Similarly, placing a saturated salt or other compatible solution in reservoir **48** will likely create a high humidity environment over the sample wells **4** sufficient to provide the desired reduction in evaporative loss from wells **4**. For example, a potassium sulfate saturated water solution is known to create a 98% humidity above the solution.

(20) Top **20** includes a downwardly projecting flange **27**. In the embodiment of FIGS. **4**, **6**, and **9** downwardly projecting flange **27** engages the outer surface **3** of sample tray **2**. In the embodiment of FIGS. **5**, **7** and **8**, downwardly projecting flange **27** further carries an outwardly projecting flange **29** suitable for supporting cover **10** or top **20** when top **20** is used alone as depicted in FIGS. **7-8**.

(21) Top **20** includes a plurality of holes **22** providing fluid communication through top **20**. As depicted in FIGS. **1-2**, **4-5**, **8**, and **9** holes **22** are arranged in a plurality of rows laid out in a rectangular fashion. Thus, holes **22** correspond in location to sample wells **4**. The plurality of rows of holes **22** are defined by a first outer perimeter row **52**, a second outer perimeter row **54**, a third outer perimeter row **56** and a fourth outer perimeter row **58**. Each outer perimeter row **52-58** shares a corner hole or location **62**, **64**, **66** and **68** with an adjacent perimeter row **52-58** as depicted in FIG. **1**. Thus, top **20** is configured to correspond to the arrangement of a conventional plate of sample tray **2**. Top **20** can be modified in configuration to accommodate alternative sample tray **2** configurations.

(22) Through multiple observations, sample wells **4** in the perimeter of multi-well sample tray **2** were determined to experience a higher rate of evaporative fluid loss than sample wells **4** to the interior of multi-well sample tray **2**. Therefore, to provide the desired evaporative control, holes **22** in top **20** have differing diameters based on their location. Holes **22** located to the interior of perimeter rows **52-58** have a first diameter ($D1$). Holes **22** within perimeter rows **52-58** have a second diameter ($D2$) which is less than the first diameter and corner holes **62**, **64**, **66** and **68** have a third diameter ($D3$). The third diameter is equal to or less than the second diameter. Thus, the diameters for each location can be stated as $D1 \geq D2 \geq D3$. The sizes of the first diameter, second diameter and third diameter will depend on the configuration of evaporation control cover **10**.

(23) When evaporation control cover **10** has a configuration similar to that of FIG. **4**, i.e., evaporation control cover **10** engages sample tray **2** and moves with sample tray **2**, then $D1$ may be between about 0.7 mm and about 5.9 mm. More typically, $D1$ may be between about 1.4 mm and about 4.8 mm. In most cases, $D1$ will be between about 1.7 mm and about 4.4 mm. In this configuration, $D2$ may be from 0.2 times $D1$ to 0.85 times $D1$ and $D3$ is from 0.15 times $D1$ to 0.7 times $D1$. For example, when $D1$ is 2.0 mm, $D2$ may be between 0.4 mm and 1.7 mm and $D3$ may be between 0.3 mm and 1.4 mm. More typically, $D2$ is from 0.3 times $D1$ to 0.8 times $D1$ and $D3$ is from 0.2 times $D1$ to 0.6 times $D1$. In most cases, $D2$ is from 0.4 times $D1$ to 0.7 times $D1$ and $D3$ is from 0.25 times $D1$ to 0.5 times $D1$.

(24) When evaporation control cover **10** has a configuration similar to that of FIG. **5**, i.e., evaporation control cover **10** is stationary with sample tray **2** moving beneath it, then $D1$ may be 0.5 mm and about 5.0 mm. More commonly, $D1$ may be between about 0.7 mm and about 4.5 mm.

More typically, D1 will be between about 0.9 mm and about 3.9 mm. In this configuration, D2 may be from 0.2 times D1 to 0.85 times D1 and D3 is from 0.15 times D1 to 0.7 times D1. For example, when D1 is 2.0 mm, D2 may be between 0.4 mm and 1.7 mm and D3 may be between 0.3 mm and 1.4 mm. More typically, D2 is from 0.3 times D1 to 0.8 times D1 and D3 is from 0.2 times D1 to 0.6 times D1. In most cases, D2 is from 0.4 times D1 to 0.7 times D1 and D3 is from 0.25 times D1 to 0.5 times D1.

(25) In some embodiments, the desired reduction in evaporation from sample wells **4** will be achieved when D2 is 50% of D1 and D3 is 33% of D1.

(26) As depicted in FIGS. 2-3, bottom **40** includes a plurality of upwardly projecting ports **42**. Ports **42** are aligned with holes **22**. Ports **42** provide fluid communication through bottom **40**. When in use, upwardly projecting ports **42** and holes **22** provide a passageway for a sensor probe **12** to pass through evaporation control cover **10** into selected sample well **4**. Additionally, as depicted in FIGS. 2, 4-5, 8, and 9 upwardly projecting ports **42** are arranged in a plurality of rows laid out in a rectangular fashion corresponding to the rows of top **20**. Thus, upwardly projecting ports **42** correspond in location to sample wells **4**. The plurality of rows of upwardly projecting ports **42** are also defined by a first outer perimeter row **72**, a second outer perimeter row **74**, a third outer perimeter row **76** and a fourth outer perimeter row **78**. Each outer perimeter row **72-78** shares a corner hole **82**, **84**, **86** and **88** with an adjacent perimeter row **72-78** as depicted in FIG. 2.

(27) Bottom **40** carries an upwardly projecting flange **44**. The region between upwardly projecting flange **44** and upwardly projecting ports **42** defines a reservoir **48**. Reservoir **48** receives the wetting fluid through port **24** or alternatively through alternative opening **26** which permits fluid to pass between top **20** and bottom **40** into reservoir **48**. Thus, liquid retained in reservoir **48** helps maintain wettable insert **30** sufficiently saturated to provide the desired evaporative control. Alternatively, as discussed above, reservoir **48** may be used to contain the desired liquid without the presence of wettable insert **30**.

(28) In some embodiments, upwardly projecting ports **42** in bottom **40** may have differing inside diameters based on their location. Upwardly projecting ports **42** located to the interior of perimeter rows **72-78** have a fourth inside diameter (D4). Upwardly projecting ports **42** within perimeter rows **72-78** have a fifth inside diameter (D5) which is less than or equal to the fourth diameter and corner holes **82**, **84**, **86** and **88** have a sixth inside diameter (D6). The sixth inside diameter is equal to or less than the fifth inside diameter. Thus, the diameters for each location can be stated as $D4 \geq D5 \geq D6$. The sizes of the fourth inside diameter, fifth inside diameter and sixth inside diameter will depend on the configuration of evaporation control cover **10**. Upwardly projecting ports **42** also have outside diameters which may be from about 1 mm to about 3 mm greater than the corresponding inside diameters. When bottom **40** has ports **42** of differing sizes as outline above, top **20** may have holes **22** of uniform diameter. Likewise, when top **20** has holes **22** of differing sizes as previously defined, then bottom **40** may have projecting ports **42** of uniform diameter.

(29) In most instances, D1 will equal D4, D2 will equal D5 and D3 will equal D6. Thus, when evaporation control cover **10** has a configuration similar to that of FIG. 4, i.e., evaporation control cover **10** engages sample tray **2** and moves with sample tray **2**, then D4 may be between about 0.7 mm and about 5.9 mm. More typically, D4 may be between about 1.4 mm and about 4.8 mm. In most cases, D4 will be between about 1.7 mm and about 4.4 mm. In this configuration, D5 may be from 0.2 times D4 to 0.85 times D4 and D6 is from 0.15 times D4 to 0.7 times D4. More typically, D5 is from 0.3 times D4 to 0.8 times D4 and D6 is from 0.2 times D4 to 0.6 times D4. In most cases, D5 is from 0.4 times D4 to 0.7 times D4 and D6 is from 0.25 times D4 to 0.5 times D4.

(30) When evaporation control cover **10** has a configuration similar to that of FIG. 5, i.e., evaporation control cover **10** is stationary with sample tray **2** moving beneath it, then D4 may be 0.5 mm and about 5.0 mm. More commonly, D4 may be between about 0.7 mm and about 4.5 mm. More typically, D4 will be between about 0.9 mm and about 3.9 mm. In this configuration, D5 may be from 0.2 times D4 to 0.85 times D4 and D6 is from 0.15 times D4 to 0.7 times D4. For example,

when D4 is 2.0 mm, D5 may be between 0.4 mm and 1.7 mm and D6 may be between 0.3 mm and 1.4 mm. More typically, D5 is from 0.3 times D4 to 0.8 times D4 and D6 is from 0.2 times D4 to 0.6 times D4. In most cases, D5 is from 0.4 times D4 to 0.7 times D4 and D6 is from 0.25 times D4 to 0.5 times D4.

(31) In some embodiments, the desired reduction in evaporation from sample wells **4** will be achieved when D5 is 50% of D4 and D6 is 33% of D4.

(32) While the embodiments of evaporation control cover **10** described above provide enhanced fluid retention and consistency across each well **4**, an alternative embodiment in which each hole **22** has identical diameters and each projecting port **42** has identical diameters will also provide enhanced fluid retention. See Table 2 below.

(33) To provide for passage of sensor probe **12** through evaporation control cover **10**, wettable insert **30** has a plurality of holes **32**. Thus, holes **32** are also arranged in the same manner as holes **22** and upwardly projecting ports **42** such that upwardly projecting ports **42** pass through holes **32**. Thus, the diameters of holes **32** correspond to the outside diameters of the corresponding upwardly projecting ports **42**.

(34) Tests were conducted to demonstrate the effectiveness of evaporation control cover **10**. Each test was conducted over a 16-hour period at 25° C. using a shaker operating at 1000 RPM. For the tests conducted using evaporation control cover **10**, wettable insert **30** is a polypropylene material with a thickness of 1.6 mm. The wetting liquid was deionized water.

(35) Table 1 serves as a control and reflects the fluid loss from wells **4** in the absence of evaporation control cover **10**. As reflected by Table 1, on average each well retained only 41.5% of the original fluid volume. The standard deviation for Table 1 is 3.2% and the coefficient of variation (% CV) is 7.6%. Table 2 reflects the improvement provided by use of evaporation control cover **10** with all holes **22** having a diameter of 3.4 mm. As reflected by Table 2, on average each well retained 93.2% of the original fluid volume. The standard deviation for Table 1 is 3.1% and the coefficient of variation (% CV) is 3.4%. Table 3 reflects the further improvement provided by using evaporation control cover **10** with varying diameter holes **22** as described above. In this instance, the outer perimeter holes (the holes within first outer perimeter row **52**, second outer perimeter row **54**, third outer perimeter row **56**, and fourth outer perimeter row **58**) have diameters of 2.4 mm, while holes **22** to the interior have diameters of 3.4 mm and holes **22** at locations **62**, **64**, **66**, **68** have diameters of 2 mm. As reflected by Table 3, on average each well retained 93% of the original fluid volume. The standard deviation for Table 1 is 1.8% and the coefficient of variation (% CV) is 2.0%. For the sake of clarity and with reference to FIG. **1** and Tables 1-3, A1 corresponds to hole **22** at location **68**, A12 corresponds to hole **22** at location **62**, H1 corresponds to hole **22** at location **66** and H12 corresponds to hole **22** at location **64**. The remaining positions in each Table correspond to holes **22** in a like manner.

(36) Thus, use of evaporation control cover **10** more than doubled the amount of fluid retained in each well **4**. While the fluid retention provided by evaporation control cover **10** with identical holes **22** and with holes **22** of differing diameters is essentially identical, the version with holes **22** of differing diameters provides the further improvement of enhanced consistency from one well **4** to another well **4**. Clearly, evaporation control cover **10** will provide a significant improvement to the evaluation of analytes as the improved fluid retention will improve confidence in the analytic results.

(37) TABLE-US-00001 TABLE 1 Control without Cover % Liquid Remaining 1 2 3 4 5 6 7 8 9 10
11 12 A 40.8 42.6 43.0 42.6 41.6 40.6 40.5 39.6 39.5 39.3 38.0 39.0 B 42.8 44.5 43.4 42.5 41.5
41.4 40.0 39.8 39.2 38.6 36.8 38.6 C 43.6 45.1 44.0 43.2 41.8 41.1 39.9 40.1 39.5 39.1 37.3 37.5 D
43.9 38.7 44.2 44.0 42.6 41.7 40.3 39.4 38.4 38.5 36.8 37.6 E 43.9 45.9 50.0 44.5 42.8 41.4 41.4
39.3 38.2 39.0 36.3 38.2 F 43.8 47.5 46.3 45.4 45.0 44.8 43.0 40.8 39.8 38.3 37.0 37.2 G 45.2 46.7
47.0 46.4 48.5 45.0 43.4 42.1 41.1 39.3 37.3 37.3 H 43.4 45.8 47.0 44.6 44.2 44.1 41.6 41.9 40.2
37.7 35.6 35.2

(38) TABLE-US-00002 TABLE 2 With Cover % Liquid Remaining 1 2 3 4 5 6 7 8 9 10 11 12 A 81.3 88.9 89.7 90.0 90.8 90.2 91.3 91.3 90.9 90.9 88.8 88.2 B 93.6 94.3 95.0 95.8 95.2 96.0 95.3 94.7 95.5 95.4 95.0 90.0 C 93.0 95.4 95.5 96.0 96.0 95.8 87.3 95.4 96.2 95.4 95.6 89.4 D 93.6 95.5 95.5 94.5 95.7 94.9 94.3 96.4 96.2 95.6 95.4 89.2 E 93.6 95.2 96.1 95.5 96.4 94.7 95.9 95.4 96.5 95.3 95.0 87.7 F 91.6 96.8 95.5 95.6 95.5 95.8 95.3 94.8 95.1 95.8 93.1 87.3 G 90.7 94.3 94.9 94.9 94.7 95.9 95.7 94.9 94.8 93.8 93.6 88.9 H 86.6 91.3 90.4 90.9 88.6 93.8 91.7 91.0 90.8 91.2 85.1 85.5

(39) TABLE-US-00003 TABLE 3 With Cover having Varying Diameter Holes % Liquid Remaining 1 2 3 4 5 6 7 8 9 10 11 12 A 93.0 93.4 92.6 91.9 91.2 91.0 90.7 88.7 87.8 86.3 87.4 87.3 B 93.1 93.1 94.6 93.5 96.9 93.7 95.8 92.5 92.4 92.3 98.0 89.7 C 91.7 95.0 92.0 94.2 93.8 94.3 93.8 94.0 93.0 92.5 90.2 90.6 D 91.3 94.1 93.9 94.0 94.3 92.8 93.8 93.4 93.3 92.9 94.1 91.0 E 90.7 94.9 94.3 94.9 94.1 93.9 94.0 93.2 92.5 93.0 94.0 92.8 F 92.3 93.7 94.0 93.3 93.5 93.5 93.3 92.8 92.6 92.9 92.2 93.5 G 91.9 93.9 94.3 93.4 93.8 93.7 94.3 93.6 93.0 93.4 93.6 93.5 H 90.7 92.4 94.0 93.9 93.7 94.7 94.8 94.6 94.2 94.0 93.9 93.8

(40) In still another embodiment, evaporation control cover **10** has a configuration similar to that of FIG. 5, i.e., evaporation control cover **10** is stationary with sample tray **2** moving beneath it. However, in the alternative embodiment depicted in FIG. 7, evaporation control cover **10** is only top **20** as this configuration lacks a bottom. In the absence of a bottom, a wettable insert is also omitted from the cover **10**. The embodiment of FIG. 7 includes the same arrangement of holes **22** discussed above and shown in FIG. 8. Specifically, holes **22** located to the interior of perimeter rows **52-58** have a first diameter (D1). Holes **22** within perimeter rows **52-58** have a second diameter (D2) which is less than the first diameter and corner holes **62, 64, 66** and **68** have a third diameter (D3). The third diameter is equal to or less than the second diameter. Thus, the diameters for each location can be stated as $D1 \geq D2 \geq D3$. The sizes of the first diameter, second diameter and third diameter will depend on the configuration of evaporation control cover **10**. In this embodiment, D1 may be 0.5 mm and about 5.0 mm. More commonly, D1 may be between about 0.7 mm and about 4.5 mm. More typically, D1 will be between about 0.9 mm and about 3.9 mm. In this configuration, D2 may be from 0.2 times D1 to 0.85 times D1 and D3 is from 0.15 times D1 to 0.7 times D1. For example, when D1 is 2.0 mm, D2 may be between 0.4 mm and 1.7 mm and D3 may be between 0.3 mm and 1.4 mm. More typically, D2 is from 0.3 times D1 to 0.8 times D1 and D3 is from 0.2 times D1 to 0.6 times D1. In most cases, D2 is from 0.4 times D1 to 0.7 times D1 and D3 is from 0.25 times D1 to 0.5 times D1.

(41) Other embodiments of the present invention will be apparent to one skilled in the art. As such, the foregoing description merely enables and describes the general uses and methods of the present invention. Accordingly, the following claims define the true scope of the present invention.

Claims

1. An evaporation control cover for use with a multi-well sample tray, said multi-well sample tray having a plurality of regularly spaced sample wells as defined by an outer perimeter of sample wells, with additional sample wells located within said outer perimeter of sample wells, said evaporation control cover comprising: a top having: a plurality of sample holes arranged to correspond to said plurality of regularly spaced sample wells and defined by an outer perimeter of sample holes with additional sample holes located within said outer perimeter of holes; said additional sample holes located within said outer perimeter of holes having a first diameter; said outer perimeter of sample holes having a second diameter; and a downwardly projecting flange carried by said evaporation control cover, said downwardly projecting flange configured to fit over said multi-well sample tray; and a wettable insert having a plurality of insert holes aligned with said sample holes, wherein said wettable insert is disposed beneath said top.

2. The evaporation control cover of claim 1, wherein said second diameter is different from said

first diameter.

3. The evaporation control cover of claim 2, wherein said second diameter is from about 0.2 times the first diameter to about 0.85 times the first diameter.

4. The evaporation control cover of claim 1, wherein said evaporation control cover is configured to fit over said multi-well sample tray, and wherein said evaporation control cover is configured to remain stationary and not inhibit movement of said multi-well sample tray.

5. The evaporation control cover of claim 1, wherein said plurality of regularly spaced sample wells are arranged in a rectangular configuration, wherein said plurality of sample holes are arranged in a rectangular configuration corresponding to said plurality of regularly spaced sample wells, wherein said plurality of sample holes are defined by a plurality of outer perimeter rows comprising a first outer perimeter row, a second outer perimeter row, a third outer perimeter row and a fourth outer perimeter row, and wherein said plurality of outer perimeter rows define said rectangular configuration as having a first corner hole, a second corner hole, a third corner hole and a fourth corner hole with additional rows of sample holes located within said rectangular configuration.

6. The evaporation control cover of claim 1, wherein the evaporation control cover is sized and shaped to allow passage of a sensor probe therethrough via the sample holes and the insert holes.

7. The evaporation control cover of claim 1, wherein the downwardly projecting flange is sized and shaped to result in a gap between the multi-well sample tray and the evaporation control cover.

8. An evaporation control cover for use with a multi-well sample tray, said multi-well sample tray having a plurality of regularly spaced sample wells defined by an outer perimeter of sample wells with additional sample wells located within said outer perimeter of sample wells, said evaporation control cover comprising: a top having: a plurality of sample holes arranged to correspond to said plurality of regularly spaced sample wells; and a downwardly projecting flange carried by said top, said downwardly projecting flange configured to fit over said multi-well sample tray; a bottom having: a plurality of upwardly projecting ports, said upwardly projecting ports configured to provide fluid communication through said bottom and said upwardly projecting ports, and said upwardly projecting ports configured to correspond to said plurality of regularly spaced sample wells; and an upwardly projecting flange carried by said bottom, said upwardly projecting flange configured to fit within said downwardly projecting flange of said top; a region disposed within said upwardly projecting flange and around said upwardly projecting ports that forms a reservoir; and a wettable insert positioned within said reservoir, wherein said upwardly projecting ports are configured to provide fluid communication between said sample holes and said regularly spaced sample wells, and wherein said top, said wettable insert, and said bottom combined together are configured to reduce or preclude the loss of liquid from said plurality of regularly spaced sample wells.

9. The evaporation control cover of claim 8, wherein said evaporation control cover is configured to fit over said multi-well sample tray, and wherein said evaporation control cover is configured to remain stationary and not inhibit movement of said multi-well sample tray.

10. The evaporation control cover of claim 8, wherein said evaporation control cover is configured to engage said multi-well sample tray.

11. The evaporation control cover of claim 8, wherein the plurality of regularly spaced sample wells of the multi-well sample tray are arranged in a rectangular configuration defined by a plurality of outer perimeter rows comprising a first outer perimeter row, a second outer perimeter row, a third outer perimeter row and a fourth outer perimeter row, wherein said plurality of outer perimeter rows defines said rectangular configuration as having a first corner well, a second corner well, a third corner well and a fourth corner well with additional rows of sample wells located within said rectangular configuration; wherein the plurality of sample holes are arranged in a rectangular configuration corresponding to said plurality of regularly spaced sample wells, wherein the plurality of sample holes are defined by a plurality of top outer perimeter rows comprising a

first top outer perimeter row, a second top outer perimeter row, a third top outer perimeter row and a fourth top outer perimeter row, and wherein said plurality of top outer perimeter rows defines said rectangular configuration as having a first corner hole, a second corner hole, a third corner hole and a fourth corner hole with additional rows of sample holes located within said rectangular configuration; and, wherein the plurality of upwardly projecting ports in the bottom are arranged in a rectangular configuration corresponding to said plurality of regularly spaced sample wells, wherein said plurality of upwardly projecting ports is defined by a plurality of bottom outer perimeter rows comprising a first bottom outer perimeter row, a second bottom outer perimeter row, a third bottom outer perimeter row and a fourth bottom outer perimeter row, and wherein said plurality of bottom outer perimeter rows defines said rectangular configuration as having a first corner upwardly projecting port, a second corner upwardly projecting port, a third corner upwardly projecting port and a fourth corner upwardly projecting port with additional rows of upwardly projecting ports located within said rectangular configuration.

12. The evaporation control cover of claim 11, wherein said sample holes within said plurality of top outer perimeter rows have a first diameter; and wherein said sample holes within said plurality of top outer perimeter rows have a second diameter which is less than or equal to said first diameter.

13. The evaporation control cover of claim 12, wherein said second diameter is from about 0.2 times the first diameter to about 0.85 times the first diameter.

14. The evaporation control cover of claim 8, further comprising a fluid port providing fluid communication through said top with said reservoir.

15. An evaporation control cover for use with a multi-well sample tray, said multi-well sample tray having a plurality of regularly spaced sample wells arranged in a rectangular configuration as defined by a plurality of outer perimeter rows comprising a first outer perimeter row, a second outer perimeter row, a third outer perimeter row and a fourth outer perimeter row, said plurality of outer perimeter rows defining said rectangular configuration as having a first corner well, a second corner well, a third corner well and a fourth corner well with additional rows of sample wells located within said rectangular configuration, said evaporation control cover comprising: a top having: a plurality of sample holes arranged in a rectangular configuration corresponding to said plurality of regularly spaced sample wells and defined by a plurality of top outer perimeter rows comprising a first top outer perimeter row, a second top outer perimeter row, a third top outer perimeter row and a fourth top outer perimeter row, said plurality of top outer perimeter rows defining said rectangular configuration as having a first top corner hole, a second top corner hole, a third top corner hole and a fourth top corner hole with additional rows of sample holes located within said rectangular configuration; said sample holes within said plurality of sample holes and located to the interior of said first top outer perimeter row, said second top outer perimeter row, said third top outer perimeter row and said fourth top outer perimeter row having a first diameter; said sample holes within said first top outer perimeter row, said second top outer perimeter row, said third top outer perimeter row and said fourth top outer perimeter row having a second diameter which is less than or equal to said first diameter; and a downwardly projecting flange carried by said top, said downwardly projecting flange configured to fit over said multi-well sample tray; a bottom having: a plurality of upwardly projecting ports, said upwardly projecting ports configured to provide fluid communication through said bottom and said upwardly projecting ports, said upwardly projecting ports being positioned in a rectangular configuration corresponding to said plurality of regularly spaced sample wells and defined by a plurality of bottom outer perimeter rows comprising a first bottom outer perimeter row, a second bottom outer perimeter row, a third bottom outer perimeter row and a fourth bottom outer perimeter row, said plurality of bottom outer perimeter rows defining said rectangular configuration as having a first corner upwardly projecting port, a second corner upwardly projecting port, a third corner upwardly projecting port and a fourth corner upwardly projecting port with additional rows of upwardly projecting ports located within

said rectangular configuration; and an upwardly projecting flange carried by said bottom, said upwardly projecting flange configured to fit within said downwardly projecting flange of said top; a region disposed within said upwardly projecting flange and around said upwardly projecting ports that forms a reservoir; and a wettable insert positioned within said reservoir, wherein said upwardly projecting ports are configured to provide fluid communication between said sample holes and said regularly spaced sample wells, and wherein said top, said wettable insert, and said bottom combined together are configured to reduce or preclude the loss of liquid from said plurality of regularly spaced sample wells.

16. The evaporation control cover of claim 15, wherein said evaporation control cover is configured to fit over said multi-well sample tray, and wherein said evaporation control cover is configured to remain stationary and not inhibit movement of said multi-well sample tray.

17. The evaporation control cover of claim 15, wherein said evaporation control cover is configured to engage said multi-well sample tray.

18. The evaporation control cover of claim 15, wherein said second diameter is from about 0.2 times the first diameter to about 0.85 times the first diameter.

19. The evaporation control cover of claim 15, wherein said first top corner hole, said second top corner hole, said third top corner hole and said fourth top corner hole together form a plurality of corner holes; wherein said plurality of corner holes have a third diameter; wherein said upwardly projecting ports located within said rectangular configuration have a fourth inside diameter; wherein said upwardly projecting ports within said plurality of bottom outer perimeter rows have a fifth inside diameter which is less than or equal to said fourth inside diameter; and wherein said first corner upwardly projecting port, said second corner upwardly projecting port, said third corner upwardly projecting port, and said fourth corner upwardly projecting port together form a plurality of corner upwardly projecting ports, and wherein said plurality of corner upwardly projecting ports have a sixth inside diameter which is less than or equal to said fifth inside diameter.

20. The evaporation control cover of claim 15, further comprising a fluid port providing fluid communication through said top with said reservoir.
